



SECOND SEMESTER 2025-26
COURSE HANDOUT

Date: 6.01.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : CS/EEE/ECE/INSTR F241
Course Title : **Microprocessors Programming and Interfacing**
Instructor-in-Charge : **Prof. Meetha V Shenoy**
Instructor(s) : **Prof. Janardan Prasad Misra**
Tutorial Instructors : **Prof. Meetha V Shenoy, Prof. GSS Chalapathi, Prof. Tejasvi Alladi, Prof. Gopal Singh Phartiyal, Prof. Govind Prasad, Mr. Balamurugan, Mr. Aniket Upadhyay,**

1. Course Description: Programmers model of processor, processor architecture; Instruction set, modular assembly programming using subroutines, macros etc.; Timing diagrams; Concept of interrupts: hardware & software interrupts, Interrupt handling techniques, Interrupt controllers; Types of Memory & memory interfacing; Programmable Peripheral devices and I/O Interfacing; DMA controller and its interfacing; Design of processor based system. This course will have laboratory component.

2. Scope and Objective of the Course: This course is a basic introduction to processor ISA, Assembly programming, Computer & Embedded Architecture. Intel 80x86 is used as a platform through the course. 8086 - 80486 Programmers model of processor, processor architecture; Instruction set, modular assembly programming using subroutines, macros etc.; Timing diagrams; Concept of interrupts: hardware & software interrupts, Interrupt handling techniques, Interrupt controllers. Types of Memory & memory interfacing. Programmable Peripheral devices and I/O Interfacing, DMA controller and its interfacing. Design of processor based system.

3. Text Books: Barry B Brey, The Intel Microprocessors. Pearson, Eight Ed. 2009.

4. Reference Books: Douglas V Hall, Microprocessor and Interfacing, TMH, Second Edition.

5. Course Plan:

Module No.	Lecture Session	Lecture Session	Reference (Text book)	Learning outcomes
1	1	Compute Architecture, Memory & I/O organization, CISC/RISC processors	Chapter 1	Learn: Introduction to Microprocessor and Microcomputers
1	2-3	Programmer's Model	Chapter 2	Learn: 8086 Microprocessor & its architecture
1	4-6	Addressing Modes	Chapter 3	Learn: Assembly Programming
1	7-14	Instruction Set & ALP	Chapter 4-6, 8	Learn: Assembly Programming



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2	15-17	Pin Out, Modes of operation, Clocking, Buses	Chapter 9	Learn: 8086/8088 Hardware Specifications
2	18-20	Memory Devices, Address Decoding- Memory Interface 8086- 80386, Protected Memory	Chapter -10	Learn: Memory Interface and Protected memory
2	21	Basic I/O interfacing (I/O mapped I/O and Memory mapped I/O) I/O port address decoding, Protected Mode	Chapter 11.1, 11.2	Learn: I/O Interfacing
2	21-24	8255	Chapter 11.3	Learn: Programmable Peripheral Devices
3	25-26	Types of interrupts, Vector tables, Priority Schemes	Chapter 12.1, 12.2,	Learn: Interrupts
3	27-28	8259	Chapter 12.4	Learn: Interrupt Controller
3	29-30	8253/8254	Chapter 11.4	Learn: Programmable Timer
3	31-32	ADC, DAC	Chapter 11.6	Learn: Converters
3	33-34	DMA controller 8237	Chapter -13	Learn: DMA controller
3	35-36	Processor-based system design	Chapter 15	Learn: System Design
3	37-38	Advanced Processors	Chapter 16	Learn: 80186-80486
3	39-40	Bus interfaces	Chapter 15	Learn- ISA, PCI, Com, USB,AGP

6. Evaluation Scheme:

Component	Duration	Weightage (%) 300 M	Date & Time	(Close Book/ Open Book)
Mid-Semester Test	90 Min.	25% (75M)	Decided by AUSGD	CB
Lab (Regular)	2 hr	4.6% (14M)	Best 7 out of 9	OB
Tutorial exams	periodic	10.6% (32M)	Best 4 out of 6	CB
Lab test	1.5hr	14% (35M)	To be announced	OB
Design Test	1.5-2 hr	8% (24M)	To be announced	OB
Comprehensive Examination	3 hr	40% (120M)	Decided by AUSGD	CB + OB

7. Chamber Consultation Hour: To be announced in the class



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- 8. Notices:** Will be shared by nalanda periodically to students.
- 9. Make-up Policy:** Make up will be allowed for genuine cases. No make up for Lab/Tutorial exams.
- 10. Note (if any):** Minimum performance requirement for a valid grade: A student should obtain 25% of the average of the class, or 10% of the course total marks, whichever is lower, to clear the course. If any student gets marks lower than the prescribed standard mentioned above, he/she may be awarded an NC report.

Instructor-in-charge
Course No. CS/EEE/INSTR/ECE F241